PHISHGUARD-PK

AI-Based Multilingual Scam and Phishing Detection System

Software Implementation and Evaluation Record | Version 2.0 | 1 November 2026

BS Computer Science Final Year Project

Based on the supplied 10-point developer requirements brief. Covers English, Roman Urdu and Urdu messages and offline URL risk analysis.

Status: Working end-to-end software, NOT a completed external scientific validation. Exact remaining requirements are documented below. This is a **draft**, not a university-approved thesis.

Executive Summary

PhishGuard-PK accepts message text through a responsive browser app, Colab-compatible Gradio UI or JSON API. A multilingual text classifier and offline URL features feed a combined risk indicator, producing Safe, Suspicious or Scam with interpretable warning signals. New in v2 are CSV batch screening (up to 100 messages), downloadable reports, comprehensive API, and external evaluation tooling.

The model has not been verified against independently reviewed, representative real-world phishing messages. A Safe classification cannot guarantee legitimacy; a Scam result is not proof of fraud. Do not interpret risk indicators as statistical fraud probabilities.

1. Requirements Traceability

#	University developer brief	Implementation / research status
1	3-language 3,000-5,000 labeled data	Development data and candidate inventory; independent label verification missing
2	NLP normalization and TF-IDF	Reusable multilingual normalization and TF-IDF implemented
3	NB, SVM, RF, XLM-R / mBERT	Three classic models trained; GPU XLM-R script and adapter supplied; training pending
4	URL features and RF risk	URL extraction/heuristics work; real labeled URL RF training remains pending
5	Combined scoring	Text and URL risk indicators implemented
6	SHAP / LIME explainability	Offline SHAP script/outputs; user-facing warnings are pattern explanations
7	REST API	FastAPI endpoints, input validation and OpenAPI included
8	Web interface	Responsive FastAPI UI and Gradio UI, batch CSV, export
9	Independent testing	Provisional holdout and 90 synthetic cases, external 50-100 real messages pending
10	README, tests, repo, diagrams	Present locally; GitHub remote and advisor sign-off pending

2. Processing Pipeline

Message input -> validation -> language detection -> Roman-Urdu/Urdu normalization -> TF-IDF text model (or optional fine-tuned XLM-R) -> passive URL lexical features -> combined scoring -> explanation -> result/export.

URL content is never opened. WHOIS metadata lookups are disabled by default. No message history database is used.

3. Implemented Software

Web application: Responsive single-message entry, sample messages, Safe/Suspicious/Scam indicator, language recognition, URL breakdown, evidence terms and downloadable JSON. CSV batch upload screens up to 100 messages and downloads a results CSV.

Backend: GET /, GET /health, GET /model-status, POST /analyze, POST /analyze/batch, plus automatic API documentation at /docs. Invalid and oversized requests are rejected.

ML: Preprocessing and a 70/15/15 grouped development split. Trained Naive Bayes, calibrated Linear SVM and Random Forest, with validation model selection, held-out development metrics, confusion matrix and SHAP analysis script.

External training hooks: XLM-R GPU fine-tuning script; real-labeled URL Random Forest trainer with host-disjoint splits; 50-100+ independent-message evaluation script with duplicate checks.

4. Reproducible Local Commands

```
pip install -r requirements.txt
```

```
python run_web.py
```

Open http://127.0.0.1:8000 and API docs at http://127.0.0.1:8000/docs.

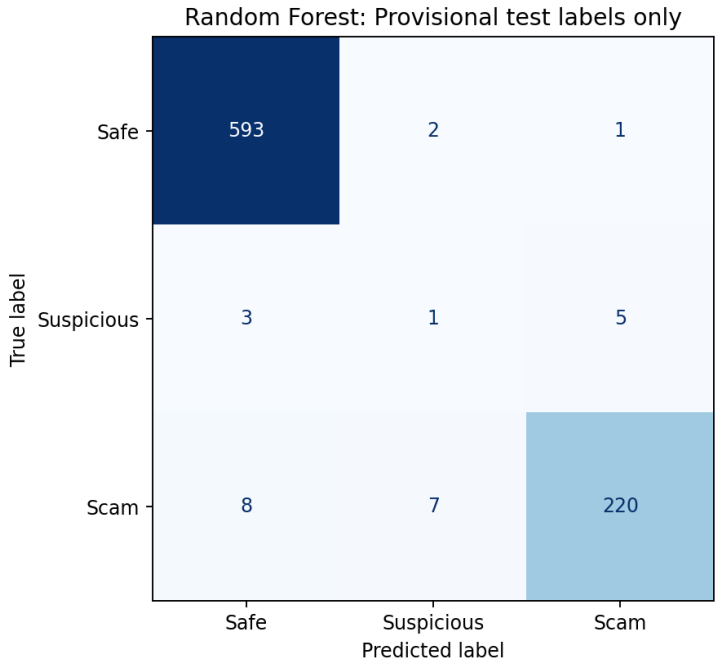
```
python -m pytest -q
```

Use the provided Colab notebook for temporary public Gradio access. Do not submit confidential text to a public Gradio share link.

5. Measured Results and Important Caveats

Evaluation	Measured result	Meaning
Software unit/API tests	32 passed	Local code and endpoint behavior, not scientific accuracy
Provisional test set	840 records	Grouped 70/15/15 development holdout; source labels weak
Validation model selection	Random Forest (macro F1 0.7649)	Calculated against provisional labels, not an external benchmark
90 synthetic regressions	59 / 90 correct verdicts	Synthetic fixtures only; highlights Suspicious-class weakness
Language detection (same 90)	87 / 90 matched	Synthetic diagnostic, not representative field accuracy

Scores from the held-out corpus and synthetic tests must NOT be described as real-world anti-phishing accuracy. Label quality, synthetic overrepresentation and template similarity materially limit conclusions.



Provisional development confusion matrix (source labels unreviewed).

6. Items Required Before Final University Sign-off

1. Independently review and approve 3,000-5,000 three-class messages. Ensure sufficient authentic Urdu-script and Suspicious samples.
2. Train XLM-RoBERTa on a GPU and benchmark it against the three classical models using the same defensible splits.
3. Obtain licensed benign/phishing URL labels, train the real URL Random Forest, and report host-independent metrics.
4. Collect and independently label 50-100 new, unseen real messages; generate per-language and class results, including error analysis.
5. Verify data licenses and personal-data handling, publish the approved source code to a GitHub repository, and get faculty approval.

7. Sources and Record of Work

Primary development reference: PhishGuard Developer Requirements, Final Year Project, BS Computer Science (10 numbered requirements, user-supplied). Development sources recorded in docs/SOURCES_AND_CREDIT.md, including UCI SMS Spam Collection, Pakistani Roman-Urdu candidates and research SMS candidates; public labels remain provisional.

URL data future reference: PhishTank official database export (https://www.phishtank.net/developer_info.php); a live feed is not bundled or claimed. Official usage conditions, licensing and any API key/rate limits must be respected.

Project implementation is delivered as versioned source code and model artifacts. The university should evaluate the working build and distinguish implementation from not-yet-completed scientific validation.